



CIS245 Project

Airplane Delay Prediction

Ayush Tripathi & Isaac Yan

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01

Objectives of the project



The Problem

Airplane delays are not just an inconvenience.



Delays cause ~\$30B in economic damages



Millions of travelers get displaced every year



Geopolitics and weather volatility increase risk

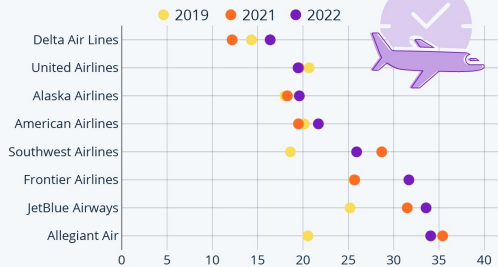


Operations depend on accurate forecasting

Sources: EDA, Statista

Delayed Flights – the New Post-Pandemic Normal?

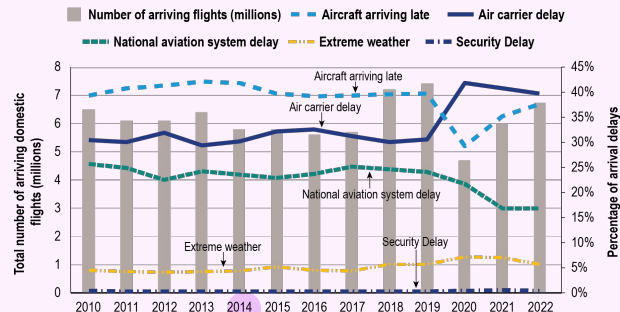
Share of late arrivals by North American airlines
(in percent of all flights)



No report for 2020. Excludes flights by partner airlines. Where data was available
Source: Cirium On-Time Performance Review



statista



Objectives

Our Aim

To develop a comprehensive forecasting system that predicts flight delays based on various factors, ranging from weather conditions to flight route information.



Key Use Cases



1. Commercial

Help passengers determine risk of delay and plan ahead of time.



2. Internal

Offer differentiated insights to airlines on what factors are most correlated with delay, allowing them to schedule more optimally.

02

Data Sources Used



Data Source 1: Weather Data

Origin & Extraction

Source: Iowa State University

Method: API Call

Scope: 3 years (top 50 US airports)

Dataset Dimensions

Raw: **1.31M Rows, 30 Features**

Cleaned: **1.31M Rows, 19 Features**

Data Cleaning Process

- Isolated date and time components for join with flight data and time series analysis.
- Created binary indicator variables to enhance model learning.
- Removed irrelevant features such as various cloud ceiling observations

Data Source 2: Flight Data

Origin & Extraction

Source: Aviation Edge API

Method: API Call

Scope: 3 years (top 50 US airports)

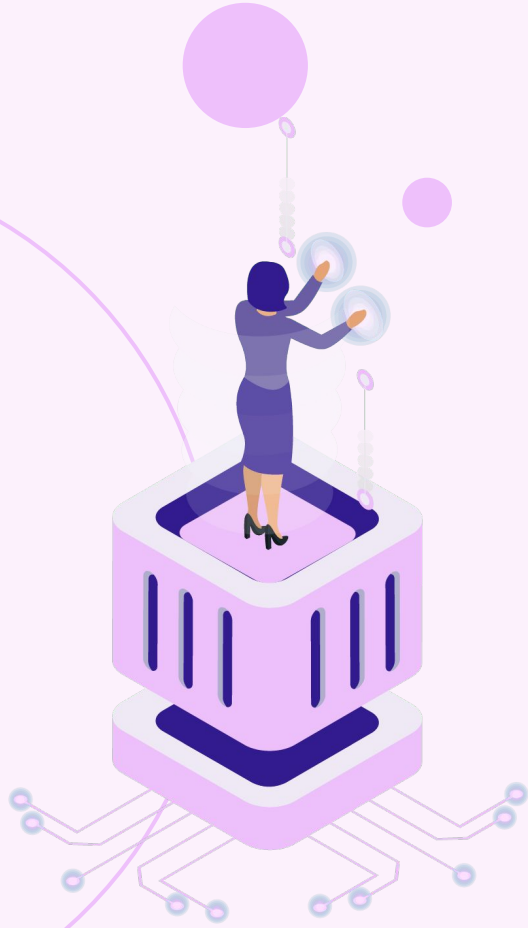
Dataset Dimensions

Raw: **20.93M Rows, 24 Features**

Cleaned: **12.16M Rows, 27 Features**

Data Cleaning Process

- Filtered out flights that were only between top 50 busiest US airports
- Added indicator variables for model categorization (delayed or not)
- Extracted date and time to join with weather data



03

Exploratory Data Analysis

EDA: Summary Statistics

Total Rows	Avg. Temp	Avg. Wind	Min. Wind	Max Wind	Avg. Visibility	Min. Visibility	Max Visibility	Max Precip.
1,313,680	63.448444	7.027962	0.0	170.0	9.497779	0.0	21,005.0	3.98

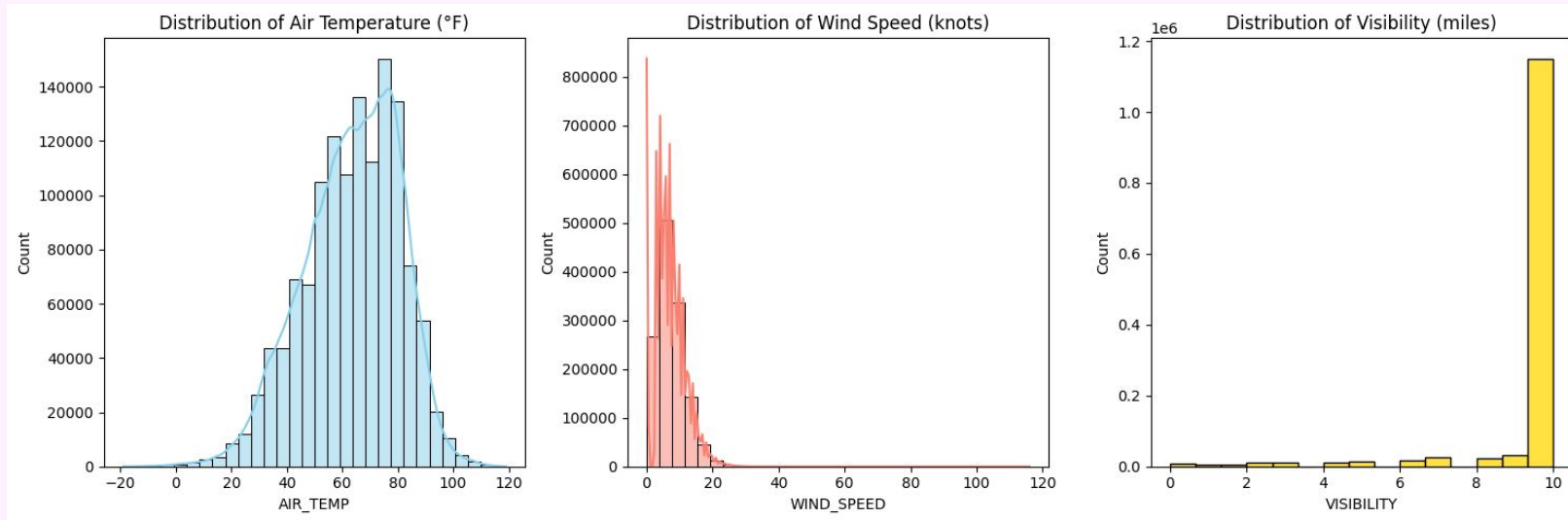
Unadjusted

Total Rows	Avg. Temp	Avg. Wind	Min. Wind	Max Wind	Avg. Visibility	Min. Visibility	Max Visibility	Max Precip.
1313680	63.448444	7.027838	0.0	116.0	9.474638	0.0	10.0	3.98

Adjusted for Outliers

Outliers (e.g., unrealistic visibility and wind values) indicate data quality issues that must be corrected before modeling, while stable averages suggest most conditions are normal. This reinforces that rare extreme weather—not typical conditions—is **likely the key driver of flight delays**, guiding our focus toward tail events rather than averages.

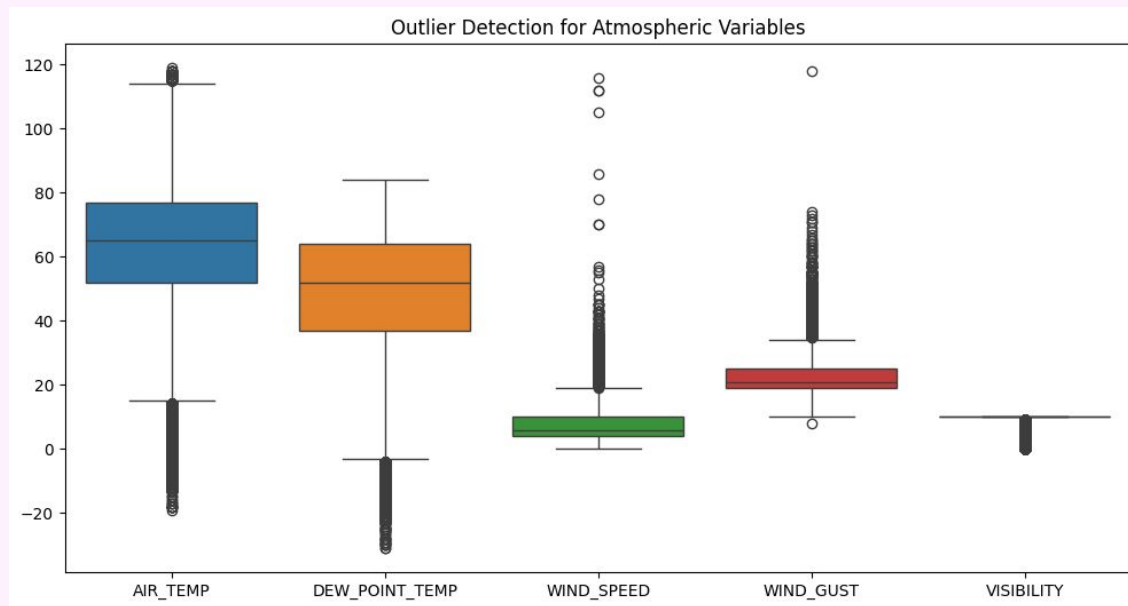
EDA #1: Feature Distribution



Distribution of air temperature, wind speed, and visibility

Temperature is roughly normal, while wind speed is right-skewed with occasional extremes. Visibility is mostly at its maximum, so **rare poor conditions likely drive delays.**

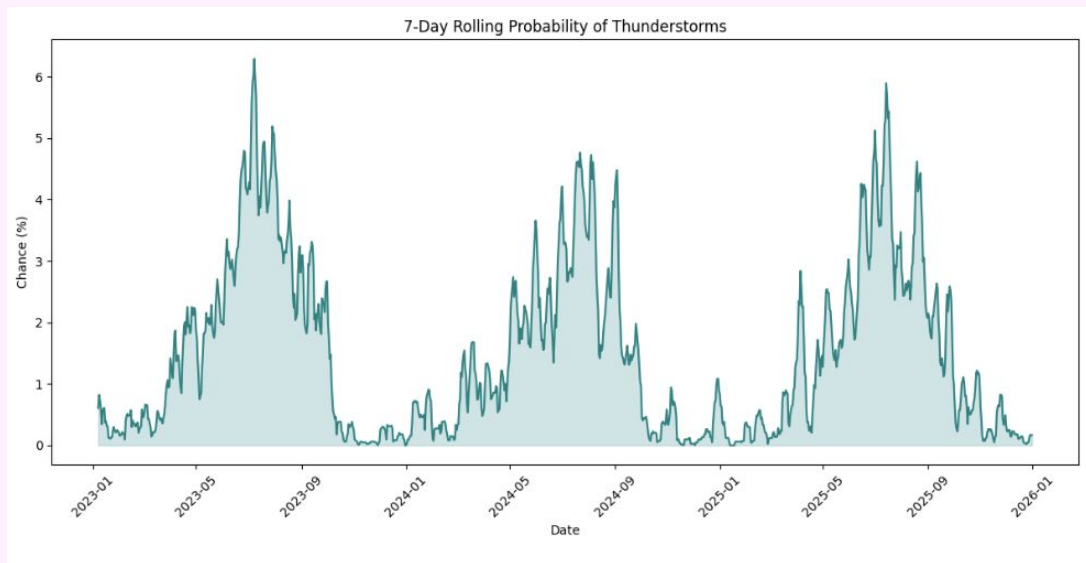
EDA #2: Outlier Detection



Outliers on Box Plot

Extreme values are prevalent across wind and temperature variables, suggesting that rare but severe weather events are critical signals for predicting flight delays rather than noise to remove.

EDA #3: Thunderstorm Feature Analysis



Thunderstorm Likelihood Time-Series

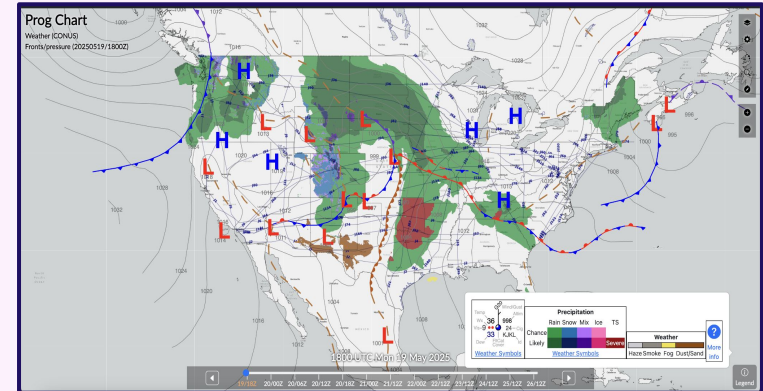
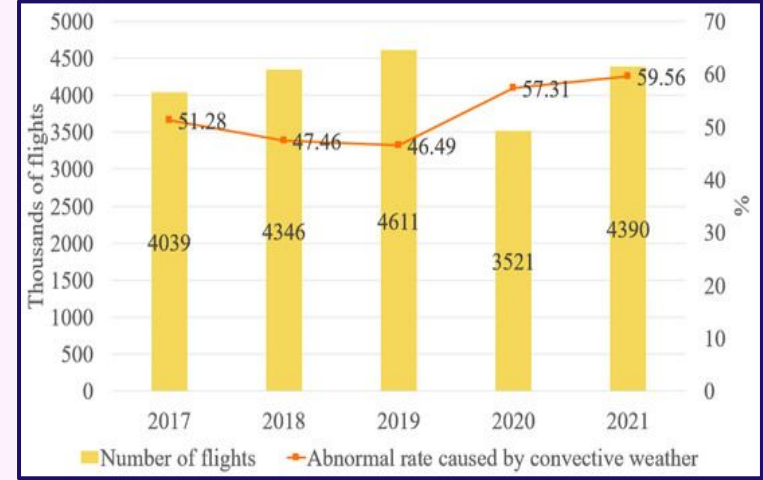
Thunderstorm probability shows strong seasonal spikes, indicating that flight delays are not random but systematically driven by predictable time-of-year patterns.

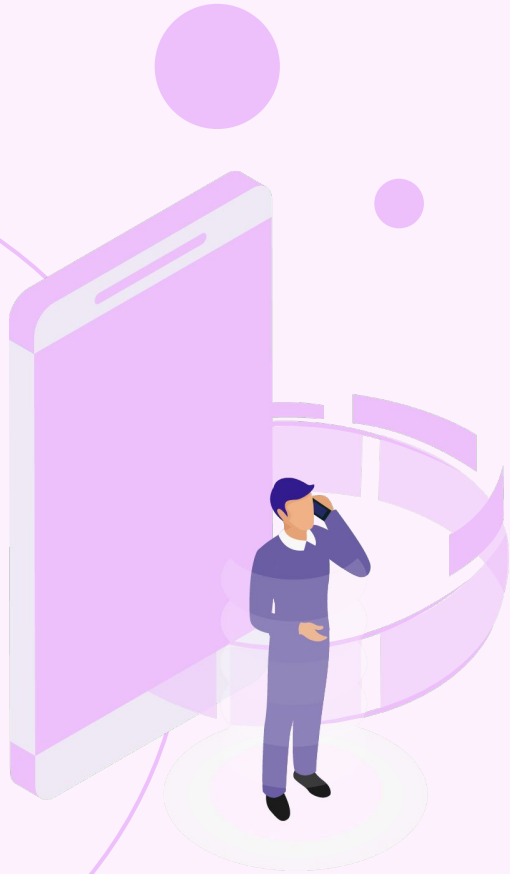
Key EDA Takeaway

EDA showed delays are driven by rare extremes.

Thus, we used non-linear models and imbalance-aware metrics to drive our research and investigation process.

The heavy tails and outliers in wind, visibility, and storm variables further indicate threshold effects and nonlinear relationships, reinforcing the need for tree-based models over linear approaches.





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Models

Modeling Approach

Problem Type

Binary classification: predict whether a flight is delayed (>15 min)

Secondary: regression on delay minutes

Challenge

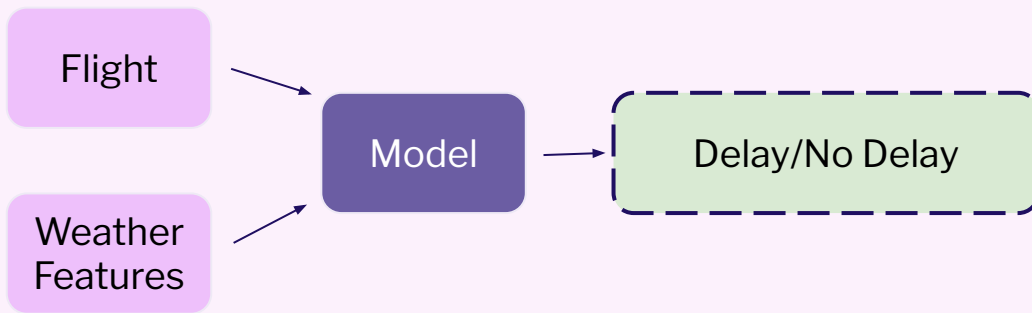
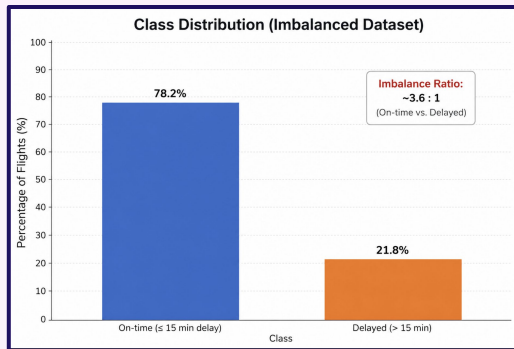
Class imbalance: most flights are on-time

Delays driven by rare extreme weather + nonlinear interactions

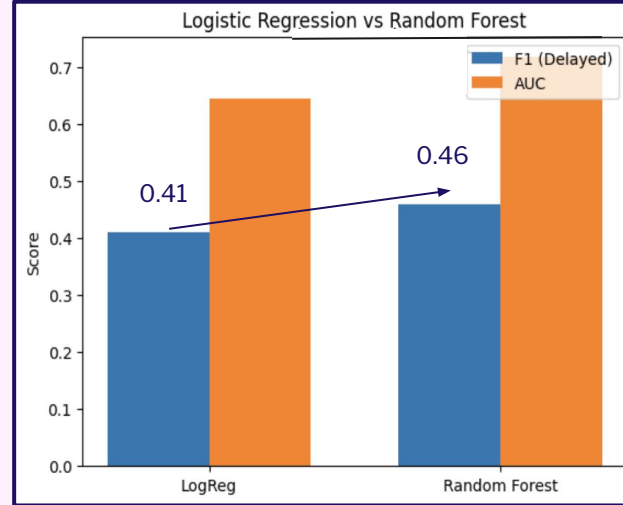
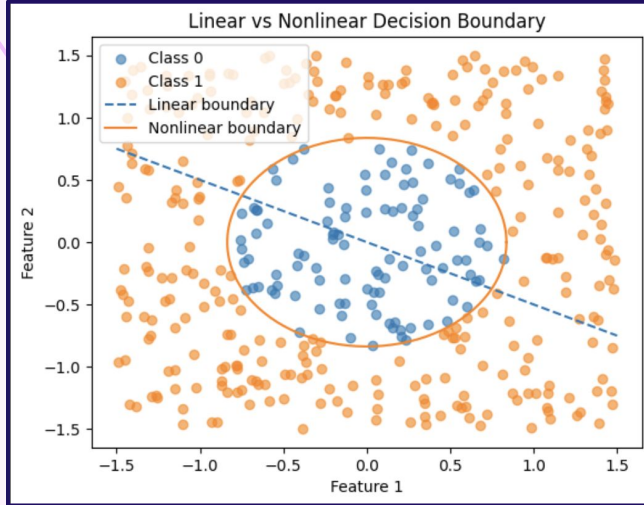
Metrics

F1 score (delayed class) and AUC

Accuracy reported but not primary due to imbalance



Linear to Non-Linear Models



Justifications

- Logistic Regression provides interpretable baseline
- Random Forest captures nonlinear interactions between features
- Significant improvement in detecting delayed flights

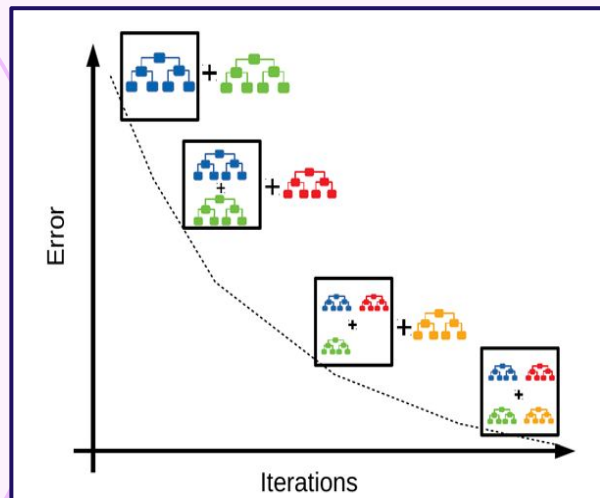
Logistic regression assumes a linear relationship between inputs and delay probability, which limits its ability to capture complex weather-flight interactions.

Random forests reduce variance through bagging while capturing higher-order feature interactions absent in linear models

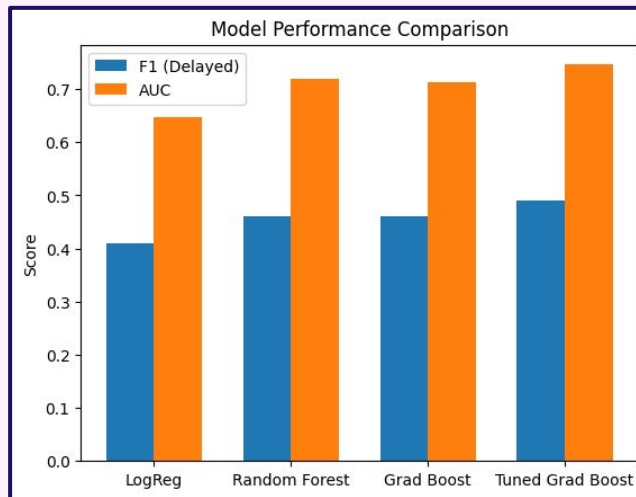
Baseline Log F1: 0.4128

Overall Accuracy: 0.579

Gradient Boosting Tree



Boosting reduces bias by sequentially fitting residual errors, which is particularly effective in imbalanced classification where minority-class errors dominate loss



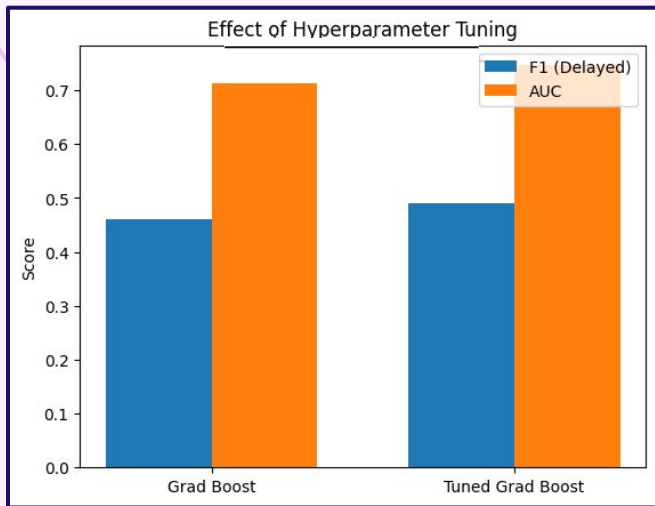
Sequential learning helps focus on hard-to-predict delayed flights, improving performance over baseline and random forest models.

Justifications

- Chosen as final model due to highest AUC and F1 score
- Better handles class imbalance and nonlinear patterns than linear models
- Hyperparameter tuning further improved generalization

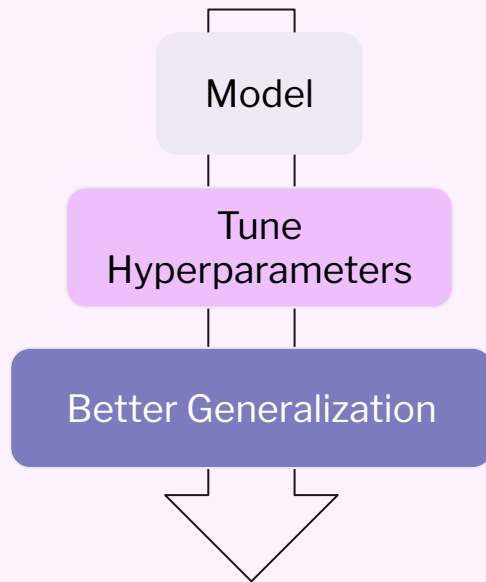
Overall F1: 0.4566
Overall Accuracy: 0.6593
AUC: 0.7126

Optimizing the Final Model



Hyperparameter tuning adjusts model complexity (e.g., tree depth, learning rate) to balance bias and variance, preventing overfitting while improving predictive performance.

Even small improvements in F1 and AUC are meaningful, as they reflect better detection of rare delayed flights—the hardest part of the problem.



Justifications

- Default parameters are not optimal for this dataset
- Tuning improves generalization to unseen data
- Leads to best overall performance across metrics

Grid/random search was used to optimize depth, learning rate, and estimators under cross-validation



05

Analysis

0.49 F1 Score

Best model performance



We optimize for F1 (delayed) because the cost of missing a delay (false negative) is higher than over-predicting one

Takeaways

Weather drives delays

Delays are strongly linked to extreme weather conditions: high wind, low visibility, storm coverage.

All increase delay probability significantly.

Delays are rare but predictable

Most flights are on time.

However, the model can still identify delayed flights by focusing on patterns in high-risk conditions.

Nonlinear models are essential

Simple linear models miss key interactions.

Tree-based models capture complex relationships between variables and improve detection of delays.

Model	F1 (Delayed)	AUC
Logistic Regression	0.41	0.65
Transformer	0.45	0.71
Random Forest	0.46	0.72
Gradient Boost	0.46	0.71
Tuned Gradient Boost	0.49	0.75

06

Other Insight



Challenges & Limitations

Class Imbalance

Severe imbalance (~78% on-time vs ~22% delayed)

Baseline accuracy is already 0.78, but F1 (delayed) is 0.00

Even our best model only reaches: 0.49 F1

The model struggles to correctly identify rare delay events, despite high overall accuracy (~0.71).

Weak Signal

Weather features provide incomplete explanatory power.

UC tops out at ~0.75, indicating only moderate separability

Delays are driven by additional unobserved factors (e.g., air traffic, airline operations, scheduling constraints) not captured in the dataset.

Precision-Recall Tradeoff

Best model metrics:

1. Precision (delayed): ~0.40
2. Recall (delayed): ~0.63

These metrics are not ideal because of one phenomenon.

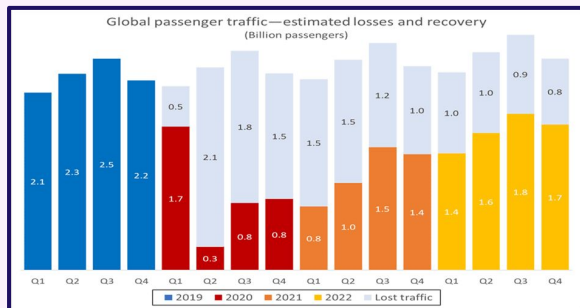
Improving recall leads to more false positives, while improving precision misses true delays—indicating an inherent tradeoff in the classification boundary.

Potential Future Work

Incorporate richer operational data

Add features beyond weather (air traffic congestion, airport capacity, airline schedules, aircraft turnaround time)

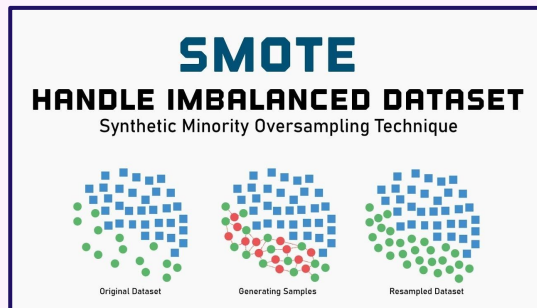
AUC suggests missing signal, so those other factors likely drive delays more directly than weather does.



Improve handling of class imbalance

Use advanced techniques like SMOTE and custom loss functions.

These methods could improve F1 scores by identifying rare delay events without sacrificing precision



Thanks!

